CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First/Second Semester of B. Tech. Examination (CE/IT/EC/EE/ME/CL)

May 2014

PY 101 Engineering Physics

Date: 23.05.2014, Friday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Choose the correct answer for the following questions. **[07]**

(a) Of these properties of a sound wave, the one that is independent of the others is its:

(i) Amplitude (ii) Frequency
(iii) Wavelength (iv) Speed

(b) A rectangular block has mass (1.5 ± 0.1) *kg*, and dimensions (80 ± 2) mm, (50 ± 1) mm and (30 ± 1) *mm*. Assuming that all the errors are independent, calculate the maximum uncertainty in the volume of the block.

(i) 5 % (ii) 10 %
(iii) 8 % (iv) 3 %

(c) Which of the following statements about sound waves is correct?

(i) The travel of sound waves is not affected by the medium through which it travels. (ii) Sound waves travel faster in air than in liquid.
(iii) Sound waves travel faster in solid than in air. (iv) Sound waves cannot travel through a solid.

(d) The higher the frequency of an ultrasound:

(i) The smaller its speed. (ii) The shorter its wavelength.
(iii) The greater its amplitude. (iv) The longer its period.

(e) ______ cables carry data signals in the form of light.

(i) Coaxial (ii) Fiber optic
(iii) Twisted pair (iv) Metallic

(f) Which of the following is not a pumping process?

(i) Optical pumping (ii) Electrical pumping
(iii) Chemical pumping (iv) Thermal pumping

(g) Production of laser does not include ________.

(i) Spontaneous emission (ii) Active medium
(iii) Stimulated absorption (iv) Population inversion

Q - 2 (a) From the length of a simple pendulum, you calculate its period of oscillation to be 0.618 ± 0.002 *s*. Experimental measurement gives 0.601 ± 0.006 *s*. Do these values agree? Explain. **[3]**

(b) A hall has a volume of 80000 cubic feet. Its total absorption is equivalent to 1000 square feet of open window. What will be the effect on the reverberation time if an audience fills the hall and thereby increases the absorption by another 1000 square feet? [4]

(c) Explain clearly what causes reverberation in a hall and how it can be minimized. What factors determine good acoustics of a room? [7]

OR

Q - 2 (a) The time period T of a simple pendulum is given by $T = 2\pi (l/g)^{1/2}$. Where l is the length of the pendulum and g is the acceleration due to gravity. If the maximum error in measuring the period of a simple pendulum is 2 % and in measuring the length is 1 %, what is the maximum percentage error in the calculated value of g? [3]

(b) State the difference between audible sound and ultrasound. Why a loud speaker cannot be used for production of ultrasonic wave? [4]

(c) What is acoustical grating? Explain how it can be used to detect and measure ultrasound. [7]

Q - 3 (a) Explain any two advantages and one disadvantage of using optical fiber. [3]

(b) Explain types of optical fiber on the basis of mode of propagation criteria. Suggest and explain which one is better for wide band communication. [4]

(c) (i) Explain the basic principle on which fiber optics works with necessary diagrams. [7]

(ii) Calculate the refractive index of glass (n_2) where numerical aperture of the air and glass interface is 0.33. Find out the critical angle and the acceptance angle both for the system.

OR

Q - 3 (a) Explain : (i) Principle of LASER, (ii) Role of active medium and (iii) Metastable state [3]

(b) (i) The ratio of two energy states is $e^{-0.5}$. Determine the temperature of the system if the lasing wavelength is 120 μm. [4]

(ii) A semiconductor material has the bandgap of 1.38 eV. Find out the wavelength in nm.

(c) What is the active medium in Nd:YAG LASER? Explain with the energy level diagram, the construction & working of Nd:YAG LASER. [7]

SECTION – II

Q - 4 **Choose the correct answer for the following questions.** [07]

(a) Below transition temperature, the electrical resistance of a superconductor is___.

(i) Finite (ii) Large

(iii) Zero (iv) None

(b) The effect of impurity on superconductor is that:

(i) T_c value increases (ii) T_c value decreases

(iii) Electrical conductivity inreases (iv) Does not change the value of T_c

(c) The intensity of X-ray can be increased in the Coolidge tube by________.

(i) By increasing the filament current (ii) By decreasing the filament current

(iii) By increasing the accelerating potential (iv) By decreasing the accelerating potential

(d) X-rays diffraction cannot be seen in plane transmission grating _____.

(i) As the slit width of the grating is greater than the wavelength of x-rays.
(ii) As the slit width of the grating is less than the wavelength of X-rays.
(iii) As the intensity of x-ray is high
(iv) both (ii) and (iii)

(e) The atomic packing factor of a face centered cubic crystal is ____.

(i) 0.52
(ii) 0.74
(iii) 0.68
(iv) 1

(f) Which of the following is not a Bravais lattice?

(i) Simple cubic
(ii) Body centered cubic
(iii) Face centered cubic
(iv) Base centered cubic

(g) If a = 10.8 Å, b = 9.47 Å, c = 5.2 Å; $\alpha = 41^0$, $\beta = 83^0$, $\gamma = 93^0$ identify unit cells into proper crystal system.

(i) Cubic
(ii) Triclinic
(iii) Hexagonal
(iv) Monoclinic

Q – 5 (a) Calculate the minimum applied potential required to produce X-rays of 1 Å wavelength. [Given, $h = 6.625 \times 10^{-34}\ J\,s$; $c = 3\times10^8\ m/s$; $e = 1.6\times10^{-19} C$]. [3]

(b) (i) Define Hall effect. [4]

(ii) An n-type semiconductor has a Hall coefficient of $3.16\times10^{-4}\ m^3C^{-1}$. The conductivity is 109 $\Omega^{-1}m^{-1}$. Calculate the carrier density n_e and the electron mobility at a room temperature.

(c) Compare Type-I and Type-II superconductors. Give examples. What are low temperature and high temperature superconductors? [7]

OR

Q - 5 (a) A zero magnetic field of a superconducting tin has a critical temperature of 3.7 *K*, the critical magnetic field is 0.306 *T*. Calculate magnetic field at 2.0 *K* [3]

(b) Explain: (i) Meissner effect, (ii) Persistent current (iii) SQUID and (iv) Critical temperature [4]

(c) Write any three applications of X-rays. A voltage of 25 *kV* is applied across the X-ray tube and a current of 10 *mA* flows through it. Find the number of electrons striking the target per second and the minimum wavelength in the X-ray spectrum. [7]

Q - 6 (a) The spacing between principal planes of *NaCl* crystal is 2.82 Å. It is found that the first order Bragg reflection occurs at an angle of 10^0. Calculate the wavelength of X-rays. [3]

(b) What are lattice parameters? How are they defined? What is their importance? [4]

(c) What is a nanostructured material? Explain briefly the effect on physical properties due to nanostructured materials. [7]

OR

Q - 6 (a) Sketch the planes for the Miller indices of (010), (001) and (200) [3]

(b) The X-ray pattern of lead with radiations 1.54 Å wavelength, the reflection from a plane with Miller indices given by (220) is observed at a Bragg's angle of 32^0. Calculate the lattice parameter of the lead. Assume first order reflection. [4]

(c) What is nanotechnology? Where is nanotechnology used today? [7]